

Customer Relationship Management (CRM) Using Agile Methodology

1. Abstract

Customer Relationship Management (CRM) is a system used by organizations to manage customer information and maintain good relationships with customers. In many organizations, customer data is still handled manually or stored in different files. This creates problems such as data duplication, data loss, and difficulty in tracking customer interactions.

The CRM system helps store customer details such as name, contact number, email address, and customer complaints in a centralized database. It also helps in managing sales records and customer support requests. By using a CRM system, organizations can improve customer satisfaction and business efficiency.

This project is developed using Agile methodology. Agile is a modern software development approach that divides the project into small phases called sprints. Each sprint includes planning, design, coding, testing, and review. Agile allows continuous improvement and makes it easy to handle changes in requirements.

The proposed CRM system is simple, user-friendly, and secure. It reduces paperwork and saves time. This system is useful for small and medium-sized organizations. In the future, more features such as mobile access, email notifications, and automation can be added.

2. Introduction

2.1 Introduction

CRM helps organizations manage customer information and improve communication with customers.

2.2 Problem Identification

- Manual data handling
- Data duplication
- Poor customer tracking

2.3 Need of the Project

- Centralized customer data
- Faster information access
- Improved customer service

2.4 Project Scheduling

The project is divided into small sprints using Agile methodology.

2.5 Objectives

- Store customer information
- Manage sales and support
- Improve customer satisfaction

3. Software Requirement Specification

3.1 Purpose

To design and develop a CRM system for managing customer data efficiently.

3.2 Scope

The system can be used by business organizations to manage customers, sales, and support activities.

3.3 Hardware / Software Requirements

Hardware

- Computer / Laptop
- Minimum 4 GB RAM

Software

- Operating System: Windows / Linux
- Programming Language: Java / Python
- Database: MySQL

3.4 Tools

- VS Code / Eclipse
- Web Browser

3.5 Software Process Model

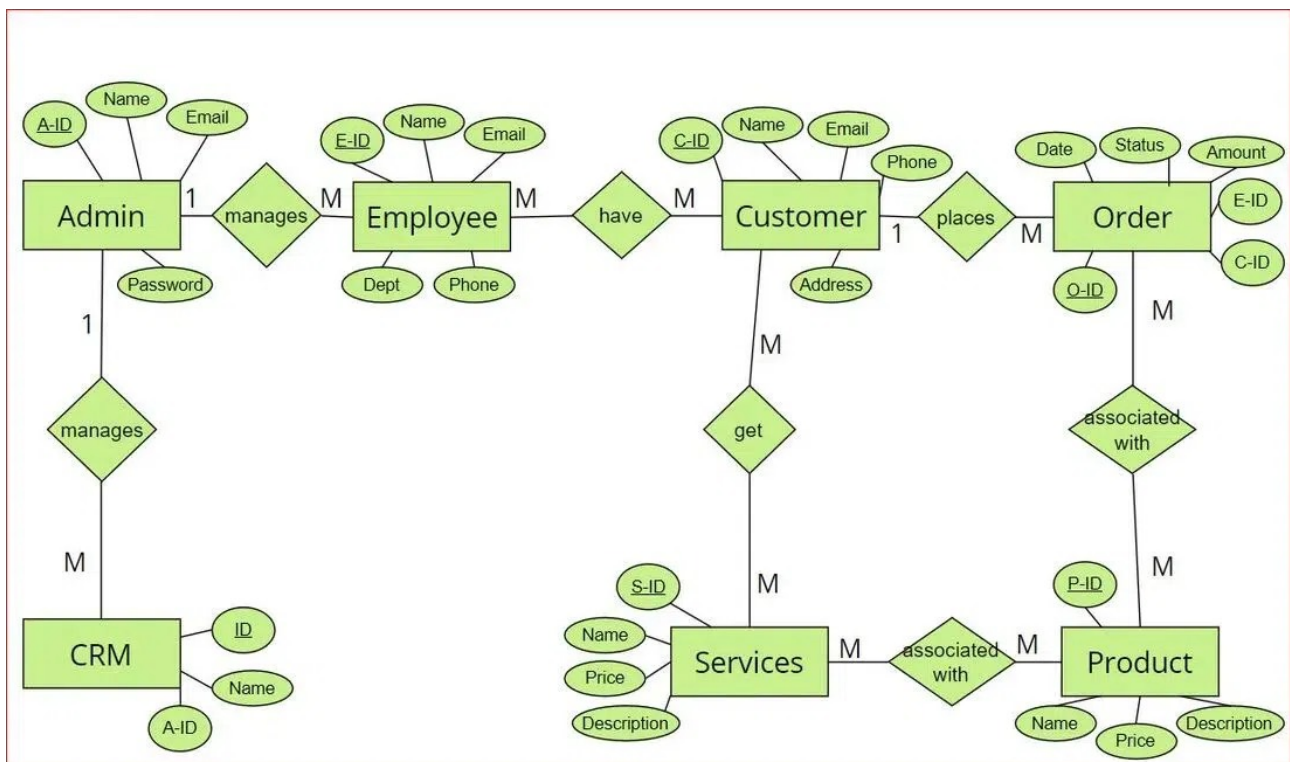
Agile Software Development Model.

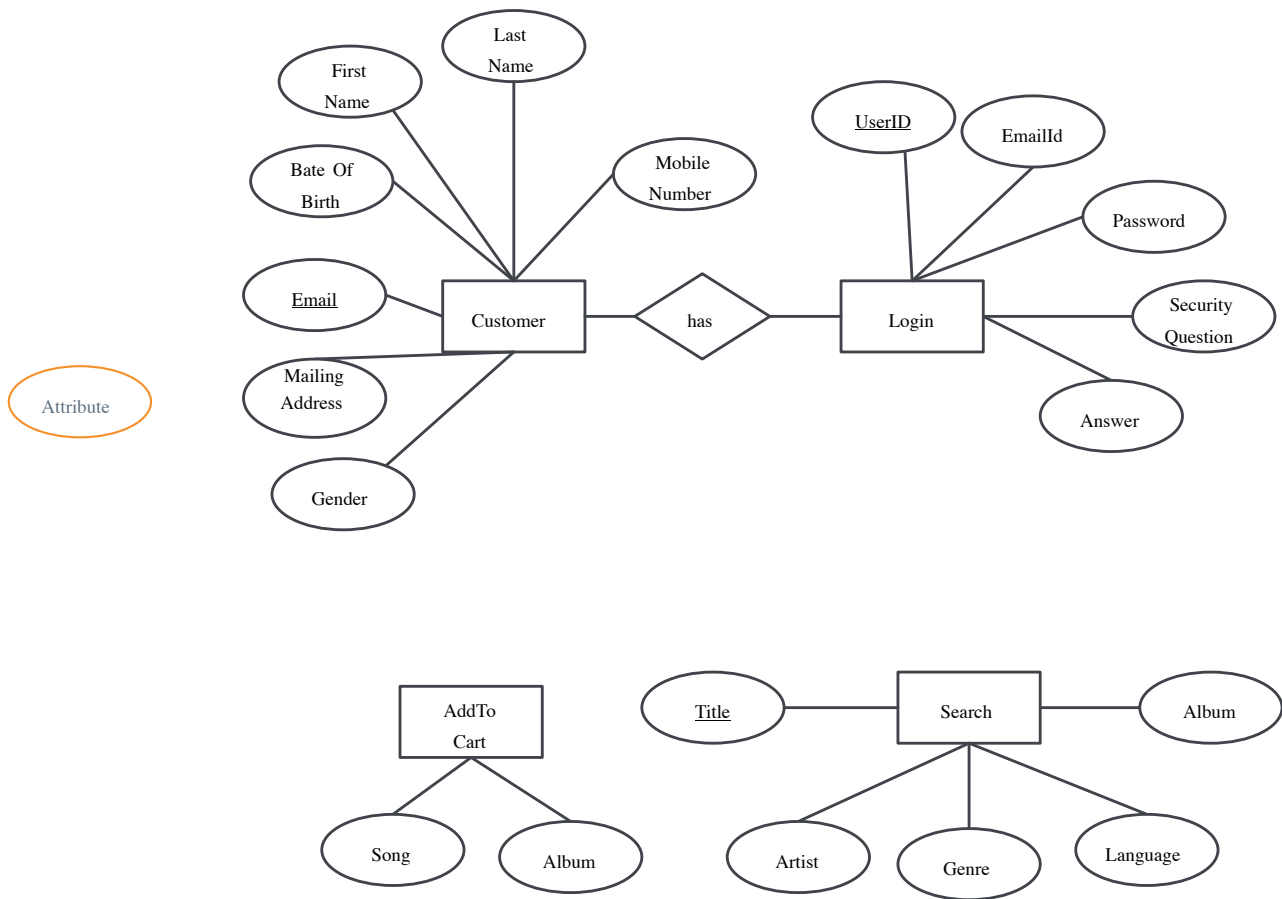
4. System Design

4.1 Data Dictionary

Field Name	Description
CustomerID	Unique identification number
Name	Name of the customer
Phone	Contact number
Email	Email address
Complaint	Customer issue or problem

4.2 ER Diagram (REAL)





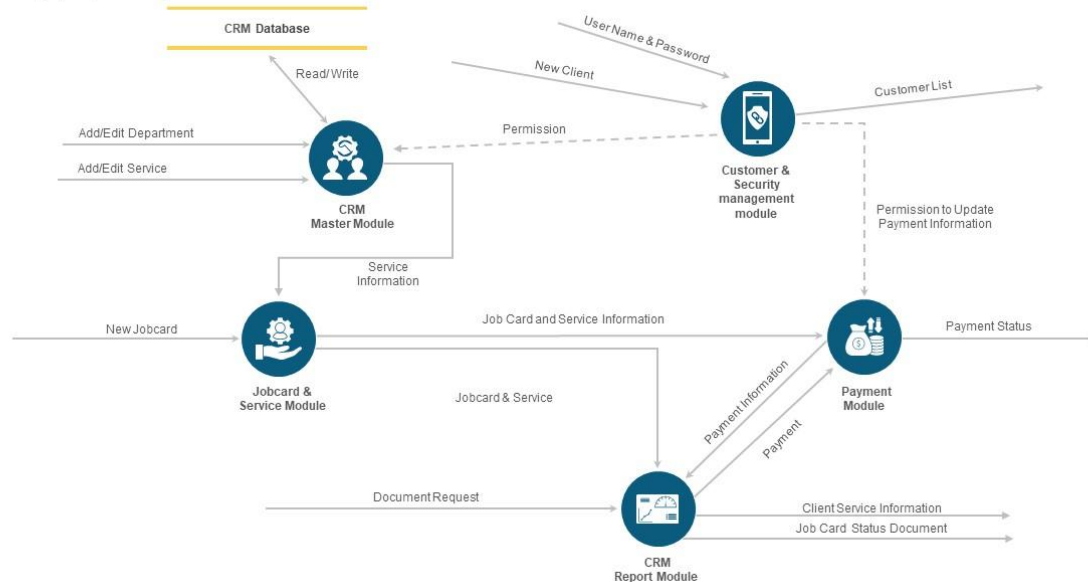
How to draw in notebook:

- Draw 3 rectangles: CUSTOMER, SALES, SUPPORT
- Write attributes inside boxes
- Connect CUSTOMER → SALES
- Connect CUSTOMER → SUPPORT
- Use straight lines only

4.3 DFD – Level 0

Customer relationship management system data flow diagram

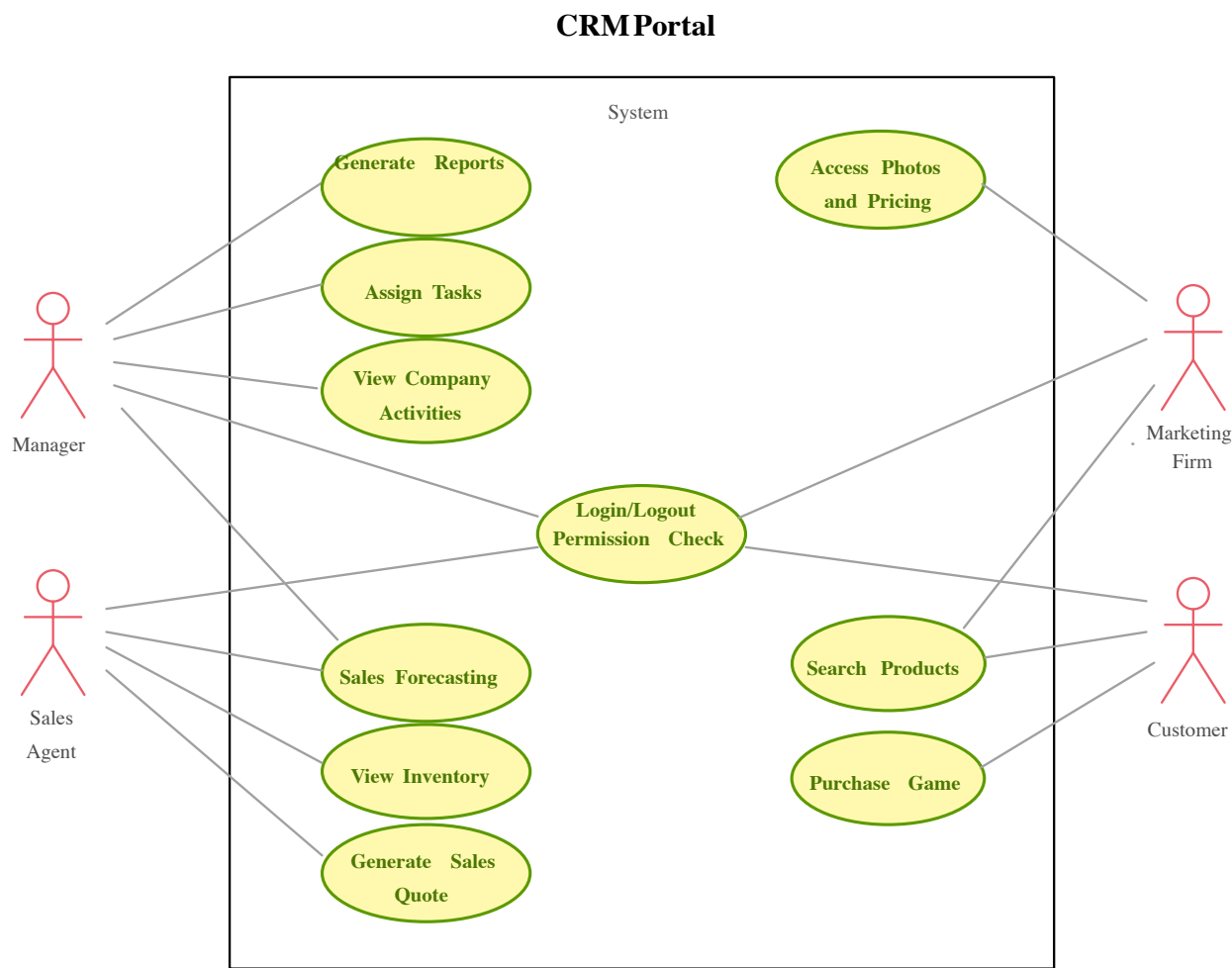
This slide displays a first-level data flow diagram for the customer relationship management system highlighting procedures used for job cards, customer registration and billing. Processes including customer & login, CRM master, job card & service, payment, and CRM reports are covered.

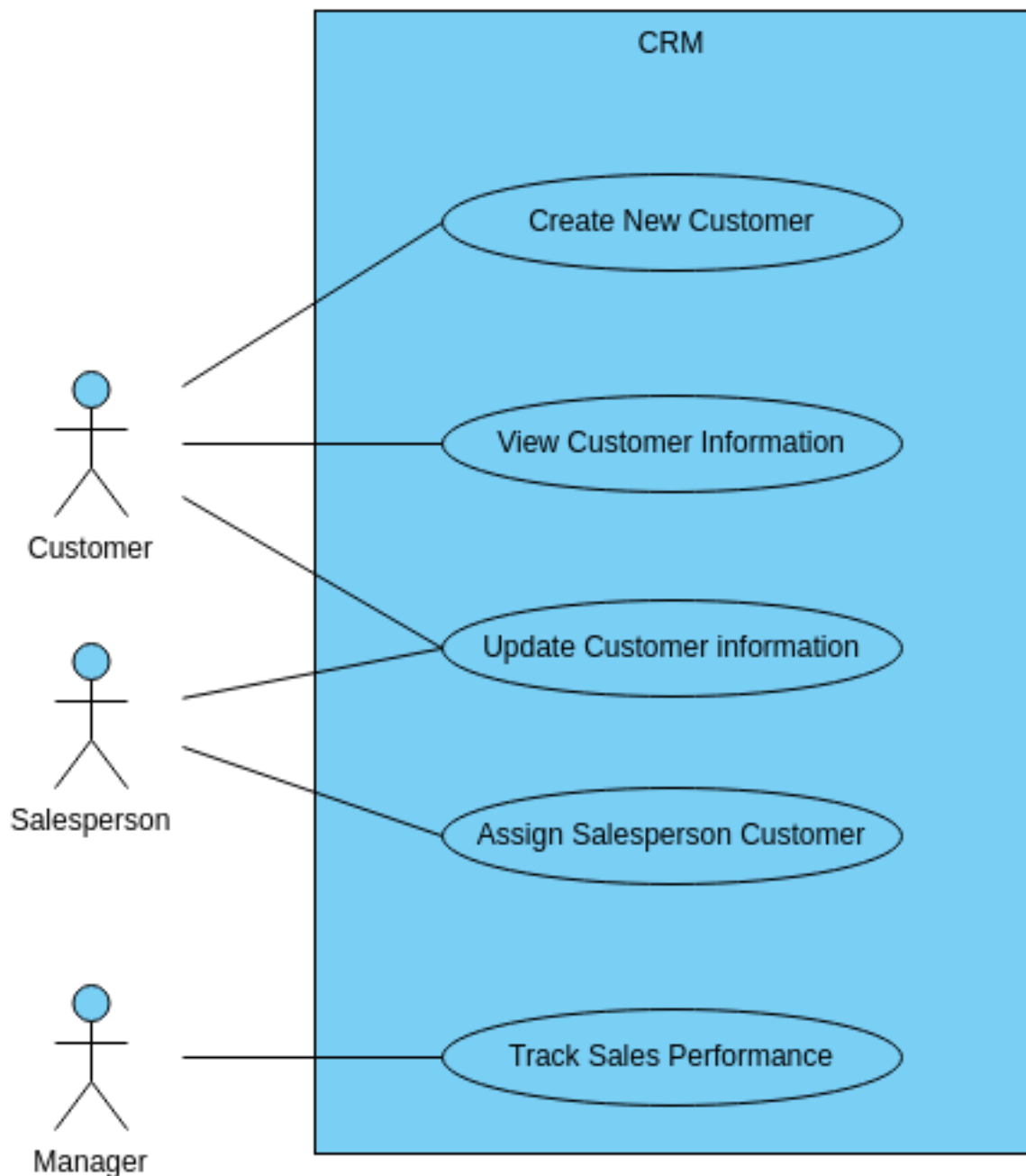


Symbols used:

- Rectangle → Customer
- Circle → CRM System
- Open-ended box → Database
- Arrows → Data Flow

4.4 Use Case Diagram





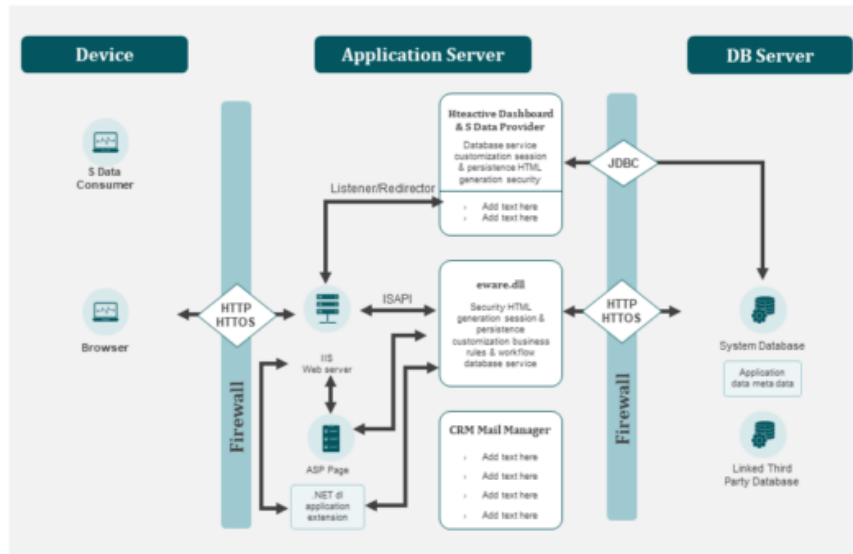
Use Cases:

- Add Customer
- View Customer
- Manage Sales
- Support Request

4.5 System Flow Chart

CRM application system architecture flowchart

This slide showcases flow diagram to help company understand CRM application architecture and working. It further includes information about device, application and DB server.



Key Features

- Interactive dashboard
- Data import and export
- Document drop panel
- Server side mail merge
- Add text here
- Add text here
- Add text here
- Add text here

This slide is 100% editable. Adapt it to your needs and capture your audience's attention.

Flow:

Start → Login → Select Module → Perform Action → Logout → End

5. Implementation

5.1 Program Code

The CRM system is implemented using Java/Python with MySQL database.

5.2 Output Screens

- Login Screen
- Customer Entry Form
- Sales Report Screen

6. Testing

6.1 Test Data

- Valid customer details
- Invalid login data

6.2 Test Result

All modules function correctly and produce expected results.

7. User Manual

7.1 How to Use Project

1. Login to system
2. Add customer details
3. Manage sales and support
4. Generate reports

7.2 Screen Layout

Simple and user-friendly interface.

8. Project Applications and Limitations

Applications

- Business organizations
- Sales departments
- Customer support centers

Limitations

- Internet required
- Basic computer knowledge needed

9. Conclusion and Future Enhancement

The CRM system helps organizations manage customer relationships effectively. Agile methodology provides flexibility and faster development. Future enhancements include mobile applications, email integration, and automated reporting.

10. Bibliography & References

- Software Engineering Textbook
- Agile Methodology Notes
- Online Reference Materials